习题课第五章

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1. 解: $\dfrac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$

$n=36$, $\mu=52$, $\sigma=6.3$

$$\frac{\bar{X}-52}{6.3/\sqrt{36}} = \frac{\bar{X}-52}{6.3/6} \sim N(0,1)$$

$$P(50.8<\bar{X}<53.8)$$

$$=P\left(\frac{50.8-52}{6.3/6}<\frac{\bar{X}-52}{6.3/6}<\frac{53.8-52}{6.3/6}\right)$$

$$=\Phi\left(\frac{53.8-52}{6.3/6}\right)-\Phi\left(\frac{50.8-52}{6.3/6}\right)$$

$=0.8302$.

$X \sim N(\mu, \sigma^2)$，　则 $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$.

$\mu = 52$，$\sigma^2 = 6.3^2$，$n = 36$.

$\therefore \bar{X} \sim N(52, \frac{6.3^2}{36})$

$P(50.8 < \bar{X} < 53.8)$

$= \Phi\left(\frac{53.8 - 52}{6.3/6}\right)$

$- \Phi\left(\frac{50.8 - 52}{6.3/6}\right)$

$= 0.8302$.

8. 解：　$X_i \sim N(0, 0.3^2)$

$$\frac{X_i - 0}{0.3} = \frac{X_i}{0.3} \sim N(0,1).$$

$$\chi^2 = \sum_{i=1}^{10} \left(\frac{X_i}{0.3}\right)^2 \sim \chi^2(10).$$

i.e.　$\underline{\frac{1}{0.09} \cdot \sum_{i=1}^{10} X_i^2 \sim \chi^2(10)}.$

$$P\left(\sum_{i=1}^{10} X_i^2 > 1.44\right)$$

$$= P\left(\underline{\frac{1}{0.09} \cdot \sum_{i=1}^{10} X_i^2} > \frac{1.44}{0.09}\right)$$

$$= P\left(\chi^2(10) > \frac{1.44}{0.09}\right)$$

$$= 0.0996.$$

12. $X \sim t(n)$，则 $X = \dfrac{Y}{\sqrt{Z/n}}$

$Y \sim N(0,1)$. $Z \sim \chi^2(n)$.

$\therefore X^2 = \dfrac{Y^2/1}{Z/n}$ $\qquad Y^2 \sim \chi^2(1)$

$\therefore X^2 \sim F(1, n)$

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13. 解：$E(\overline{X}) = E\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n} \cdot n \times E(X)$

$= E(X) = n$.

$D(\overline{X}) = D\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right)$

$= \frac{1}{n^2} \cdot D\left(\sum_{i=1}^{n} X_i\right)$

$= \frac{1}{n^2} \times n \times D(X)$

$= \frac{D(X)}{n} = \frac{2n}{n} = 2$.

15. 解：$\underbrace{X_1-2X_2}\sim N(0,5)$

$E(X_1-2X_2)=E(X_1)-2E(X_2)=0.$

$D(X_1-2X_2)=D(X_1)+4\cdot D(X_2)=5.$

$X_3+3X_4\sim N(0,10)$

$X=a(X_1-2X_2)^2+b(X_3+3X_4)^2$

$$\frac{X_1-2X_2-0}{\sqrt{5}}\sim N(0,1)$$

$$\frac{X_3+3X_4-0}{\sqrt{10}}\sim N(0,1)$$

$$\left(\frac{X_1-2X_2}{\sqrt{5}}\right)^2+\left(\frac{X_3+3X_4}{\sqrt{10}}\right)^2$$

$$\sim\chi^2(2)$$

$\therefore a=\frac{1}{5},\ b=\frac{1}{10},\ n=2$

18. 求 $\frac{X_{n+1}-\bar{X}}{S}\cdot\sqrt{\frac{n}{n+1}}$ 的分布.

$\bar{X}\sim N(\mu,\frac{\sigma^2}{n})$, $X_{n+1}\sim N(\mu,\sigma^2)$

$X_{n+1}-\bar{X}\sim N\left(E(X_{n+1}-\bar{X}),\ D(X_{n+1}-\bar{X})\right)$

$E(X_{n+1}-\bar{X})=E(X_{n+1})-E(\bar{X})=\mu-\mu=0.$

$$D(X_{n+1}-\bar{X})=D(X_{n+1})+D(\bar{X})$$
$$=\sigma^2+\frac{1}{n}\cdot\sigma^2=\frac{n+1}{n}\cdot\sigma^2$$

$\therefore X_{n+1}-\bar{X}\sim N(0,\ \frac{n+1}{n}\sigma^2)$

$$\frac{X_{n+1}-\bar{X}}{\sqrt{\frac{n+1}{n}}\cdot\sigma}=\sqrt{\frac{n}{n+1}}\cdot\frac{X_{n+1}-\bar{X}}{\sigma}$$
$$\sim N(0,1)$$

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1).$$

$$\therefore \quad \frac{\sqrt{\frac{n}{n+1}} \cdot \frac{X_{n+1}-\bar{X}}{\sigma}}{\sqrt{\frac{(n-1)S^2}{\sigma^2} \Big/ (n-1)}}$$

$$\sim t(n-1)$$

$$\therefore \quad \frac{X_{n+1}-\bar{X}}{S} \cdot \sqrt{\frac{n}{n+1}} \sim t(n-1).$$